DO NOW — RECALL

Lesson 145 · Packet Switching, Circuit Switching and Re

Name: _____ Class: _____ Date: _____

1. What is the job of DNS?

2. What is the accumulator for?

3. Which is faster in the worst case on sorted arrays, linear or binary search?

4. Which logic gate outputs 1 only when both inputs are 1?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Map names such as example.org to relevant resource records including IP addresses.

2. Model answer

Holding a CPU's working data/ALU results.

3. Model answer

Binary search: O(log n) rather than O(n).

4. Model answer

AND.

By the end, you can…

• Large message split into addressed packets for shared links.

• Packets share capacity and may arrive out of order.

• TCP can reorder and retransmit data, while IP itself is best-effort.

Vocabulary — know these terms

packet switching — Data divided into packets that may take different network routes.

circuit switching — A dedicated end-to-end circuit established before communication.

latency — Delay experienced by data in transit.

retransmission — Sending data again when loss is detected by a reliable protocol.

NEW LEARNING

Read, annotate and explain

Packetise data

Large message split into addressed packets for shared links.

Route over network

Packets share capacity and may arrive out of order.

Reassemble reliably

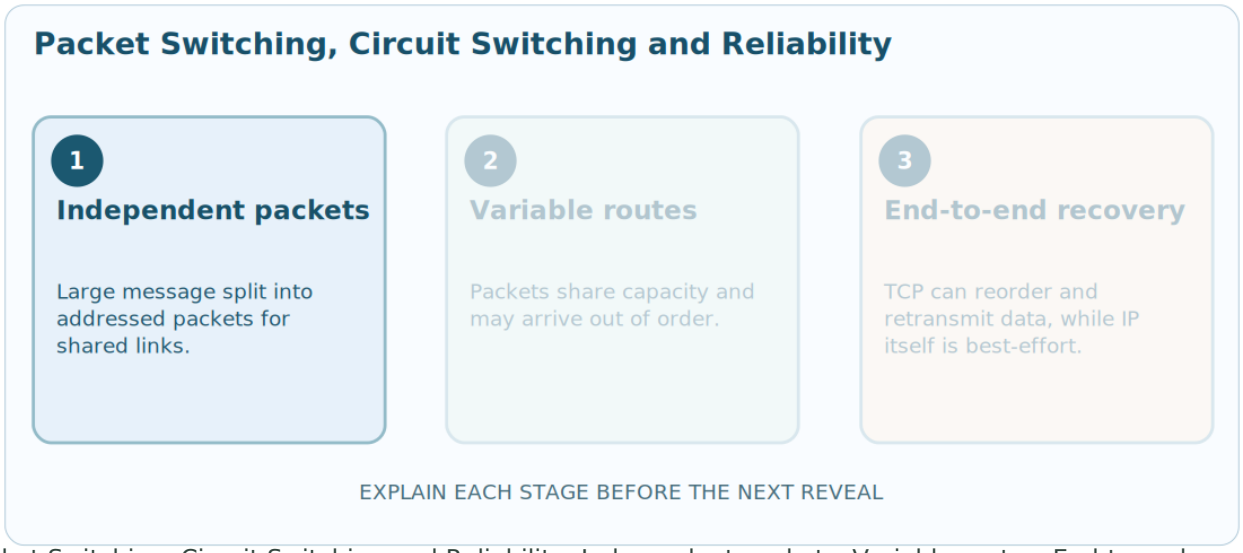
TCP can reorder and retransmit data, while IP itself is best-effort.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

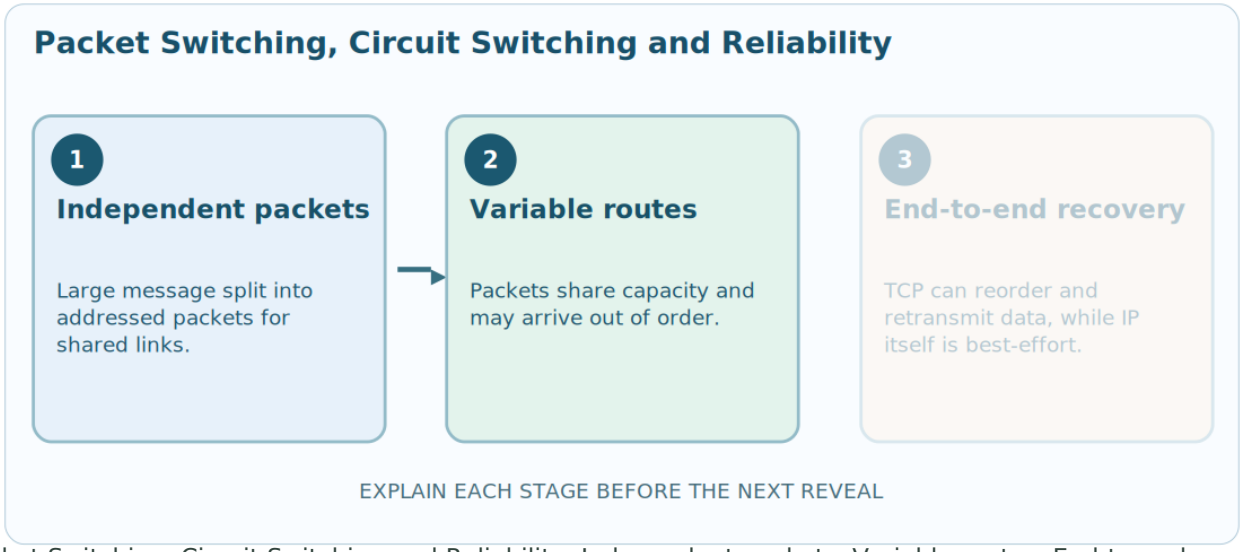
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Packet Switching, Circuit Switching and Reliability: Independent packets, Variable routes, End-to-end recovery

New Learning

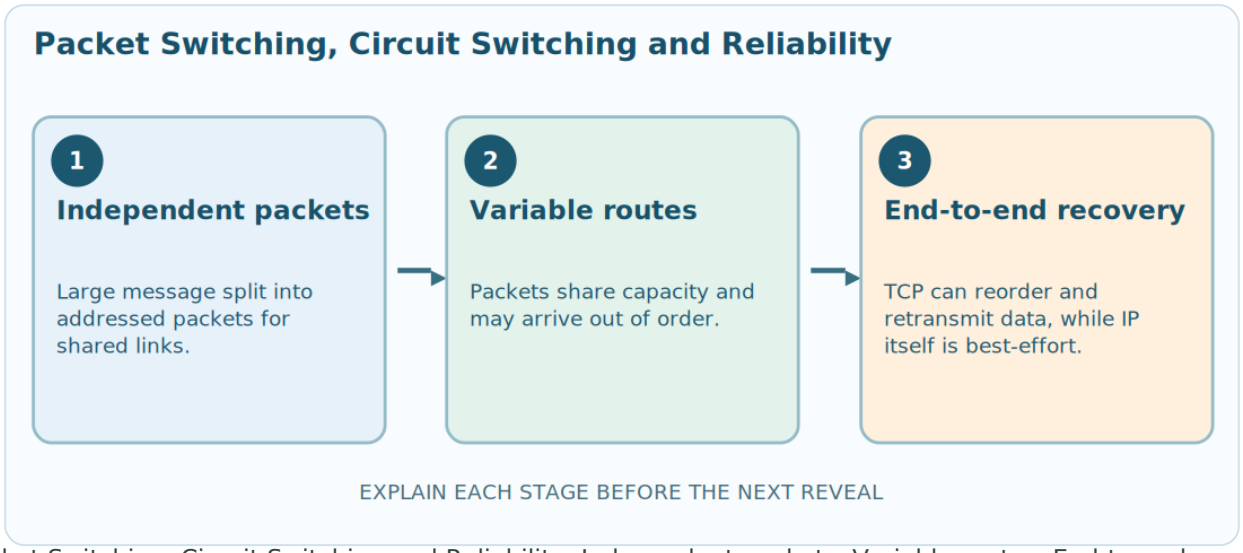


Packet Switching, Circuit Switching and Reliability: Independent packets, Variable routes, End-to-end recovery

New Learning · original lesson diagrams

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New Learning



Packet Switching, Circuit Switching and Reliability: Independent packets, Variable routes, End-to-end recovery

I DO — WORKED EXAMPLE

Follow the worked method

A 3-packet file crosses a packet-switched network. Packets arrive 2,1,3. Explain handling.

Send packets 1,2,3

Identify the relevant information and make a prediction.

Receive 2,1,3

Explain the change using correct technical terms.

Reorder 1,2,3

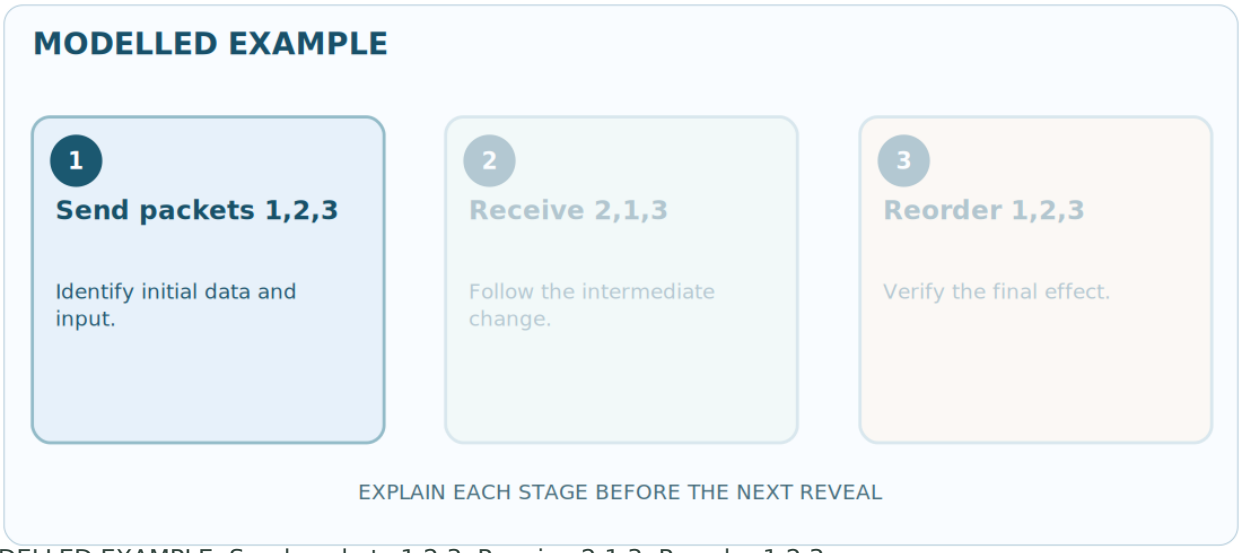
Packets can follow different routes or encounter different queue delays. With a reliable transport such as TCP, sequence numbers permit correct reordering before delivery of the byte stream; missing data can be retransmitted.

I Do · original lesson diagrams

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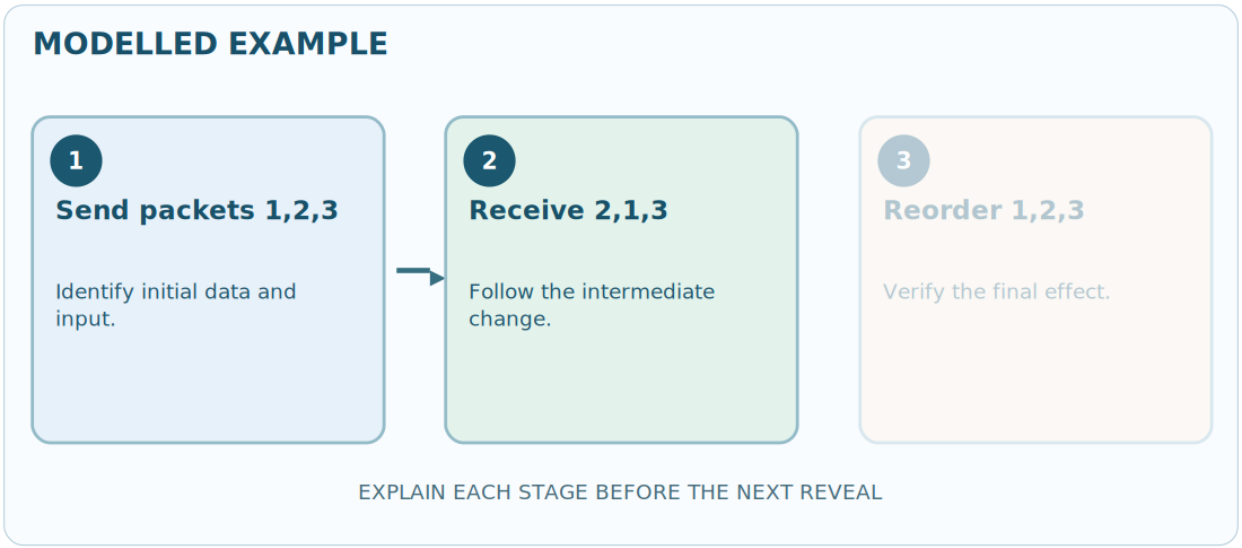
Identify the relevant information and make a prediction.



MODELLED EXAMPLE: Send packets 1,2,3, Receive 2,1,3, Reorder 1,2,3

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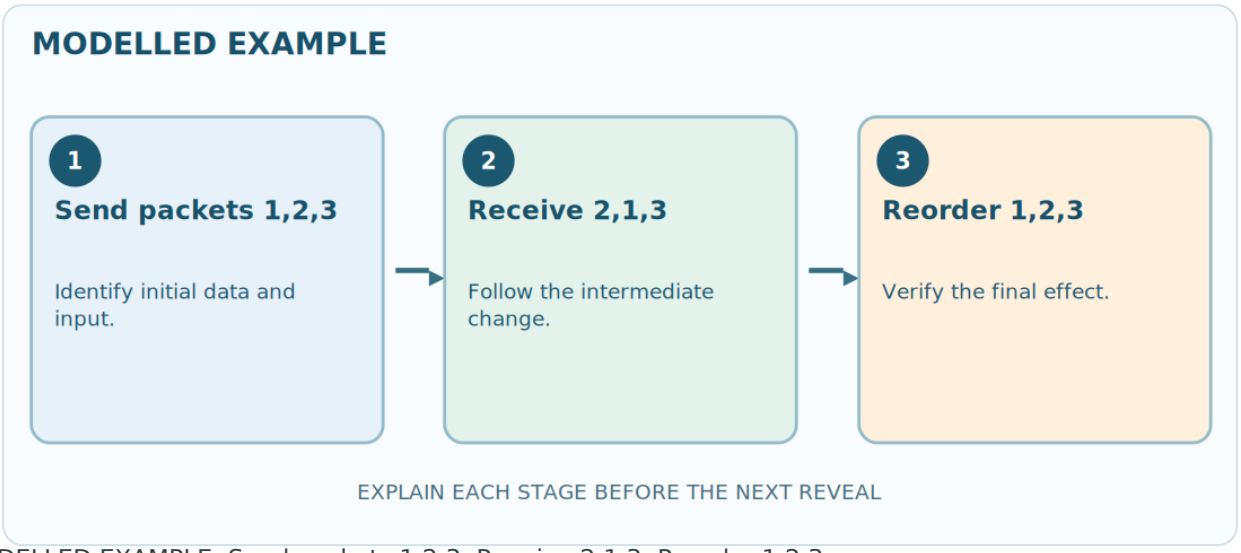
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WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A telephone system reserves capacity for a call before any voice data flows. Compare to ordinary IP networking

Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Reserve dedicated circuit, Hold resources during call, Compare shared packet links

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Reserve dedicated circuit Identify the relevant information and make a prediction.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define packet switching and circuit switching, highlighting two differences. [6]

Question 2

Explain why packets in a message may reach the destination out of order. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate packet switching versus circuit switching for video calls, including reliability and resource efficiency. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

IP alone guarantees that every packet is received in the original order.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

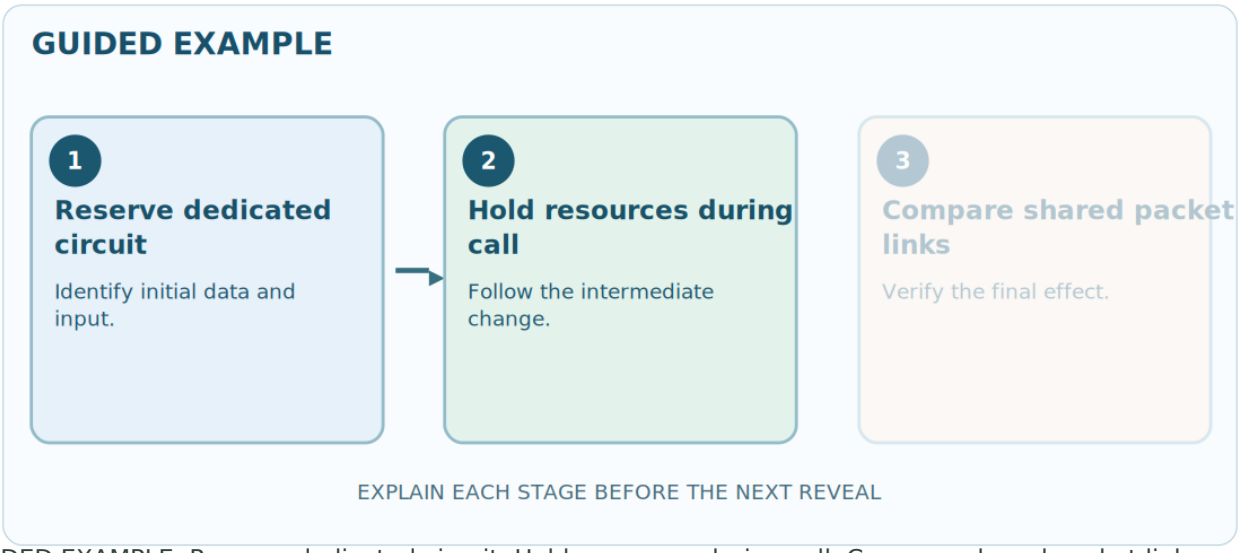
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

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A telephone system reserves capacity for a call before any voice data flows. Compare to ordinary IP networking

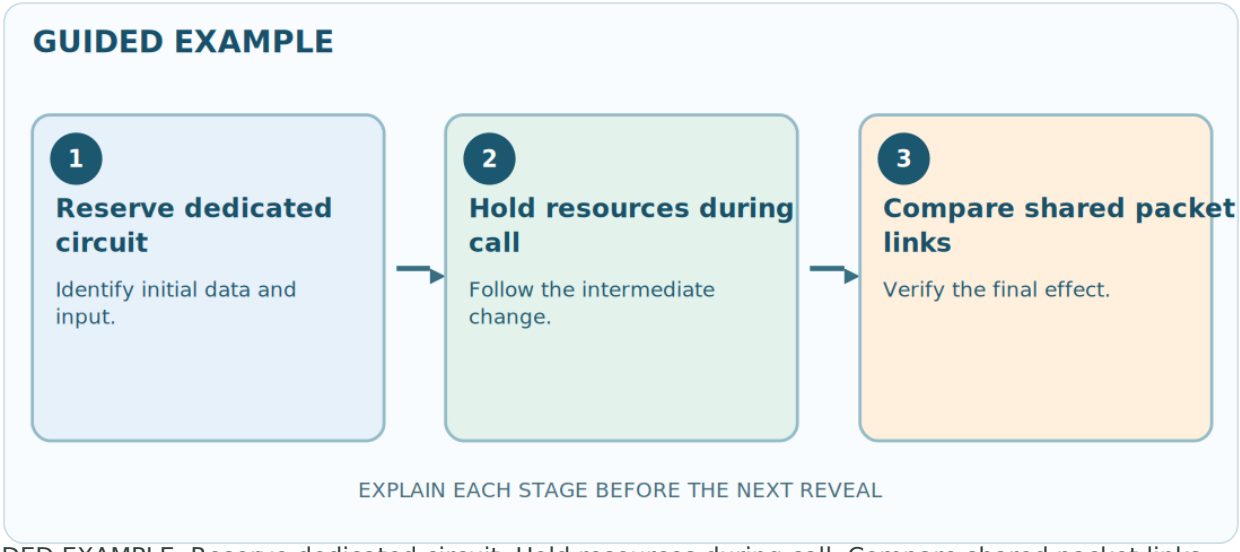
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GUIDED EXAMPLE: Reserve dedicated circuit, Hold resources during call, Compare shared packet links

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GUIDED EXAMPLE: Reserve dedicated circuit, Hold resources during call, Compare shared packet links

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Hold resources during call

Explain the change using correct technical terms.

Compare shared packet links

A circuit-switched system sets aside end-to-end resources while active; ordinary packet-switched IP shares links dynamically with others. Circuit switching offers predictable reserved capacity but can waste it during silence; packet switching adapts.

IMPROVE: Explain one correction to your own reasoning.